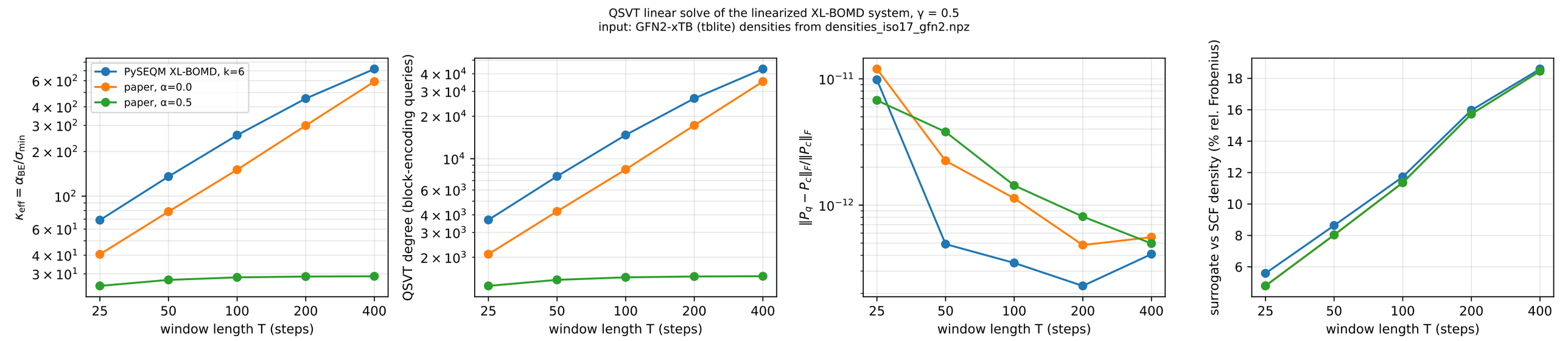
QSVT linear solve of the linearized XL-BOMD system, γ = 0.5
input: GFN2-xTB (tblite) densities from densities_iso17_gfn2.npz
PySEQM XL-BOMD, k=6
paper, α=0.0
paper, α=0.5
$\kappa_{\mathrm{eff}} = \alpha_{\mathrm{BE}}/\sigma_{\min}$
QSVT degree (block-encoding queries)
$\|P_q - P_c\|_F / \|P_c\|_F$
surrogate vs SCF density (% rel. Frobenius)
window length T (steps)
25
50
100
200
400